

Reasoning modelling

Example 1

H_1 . Every child loves *Santa*.

H_2 . Everyone who loves *Santa* loves any reindeer.

H_3 . *Rudolph* is a reindeer, and *Rudolph* has a red nose.

H_4 . Anything which has a red nose is weird or is a clown.

H_5 . No reindeer is a clown.

H_6 . *Scrooge* does not love anything which is weird.

C . *Scrooge* is not a child.

$H_1, H_2, H_3, H_4, H_5, H_6 \vdash^? C$

D (domain) = the set of all living beings

R (*Rudolf*), Sa (*Santa*), Sc (*Scrooge*) are constants from D

unary predicates:

ch (child), re (reindeer), rn (red nose), w (weird), cl (clown)

Binary predicate:

$loves$

$H_1 : (\forall x)(ch(x) \rightarrow loves(x, Sa))$

$H_2 : (\forall x)(\forall y)(loves(x, Sa) \wedge re(y) \rightarrow loves(x, y))$

$H_3 : re(R) \wedge rn(R)$

$H_4 : (\forall z)(rn(z) \rightarrow w(z) \vee cl(z))$

$H_5 : (\forall s)(re(s) \rightarrow \neg cl(s))$

$H_6 : (\forall t)(w(t) \rightarrow \neg loves(Sc, t))$

$C : \neg ch(Sc)$



Inference rules

$(\forall x)U(x) \vdash_{univ_inst} U(t)$,

t is a term (variable or constant of the domain)

$U(x) \vdash_{univ_gen} (\forall x)U(x)$,

x is a free variable in U

$(\exists x)U(x) \vdash_{exist_inst} U(c)$,

c is a new constant of the domain

$U(t) \vdash_{exist_gen} (\exists x)U(x)$,

t is a variable or a constant of the domain

$U, U \rightarrow V \vdash_{mp} V$ (modus ponens)

$U \rightarrow V, V \rightarrow Z \vdash_{syllogism} U \rightarrow Z$

$U \rightarrow V \vdash_{mt} \neg V \rightarrow \neg U$ (modus tollens)

$\neg V, U \rightarrow V \vdash_{mt} \neg U$

$H_1 : (\forall x)(ch(x) \rightarrow loves(x, Sa))$

$H_2 : (\forall x)(\forall y)(loves(x, Sa) \wedge re(y) \rightarrow loves(x, y))$

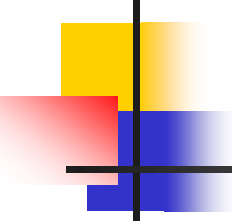
$H_3 : re(R) \wedge rn(R)$

$H_4 : (\forall z)(rn(z) \rightarrow w(z) \vee cl(z))$

$H_5 : (\forall s)(re(s) \rightarrow \neg cl(s))$

$H_6 : (\forall t)(w(t) \rightarrow \neg loves(Sc, t))$

$C : \neg ch(Sc)$



Linear resolution

$$H_1 : (\forall x)(ch(x) \rightarrow loves(x, Sa))$$

$$H_2 : (\forall x)(\forall y)(loves(x, Sa) \wedge re(y) \rightarrow loves(x, y))$$

$$H_3 : re(R) \wedge rn(R)$$

$$H_4 : (\forall z)(rn(z) \rightarrow w(z) \vee cl(z))$$

$$H_5 : (\forall s)(re(s) \rightarrow \neg cl(s))$$

$$H_6 : (\forall t)(w(t) \rightarrow \neg loves(Sc, t))$$

$$C : \neg ch(Sc)$$

Example 1 (contd.)



We apply a refutation proof method: general predicate resolution.
The clausal normal forms corresponding to the hypotheses and the negation of the conclusion are as follows:

$$H_1^C : \neg \text{child}(x) \vee \text{loves}(x, \text{Santa}) = C_1$$

$$H_2^C : \neg \text{loves}(x, \text{Santa}) \vee \neg \text{reindeer}(y) \vee \text{loves}(x, y) = C_2$$

$$H_3^C : \text{reindeer}(\text{Rudolf}) \wedge \text{red_nose}(\text{Rudolf}) = C_3 \wedge C_4$$

$$H_4^C : \neg \text{red_nose}(z) \vee \text{weird}(z) \vee \text{clown}(z) = C_5$$

$$H_5^C : \neg \text{reindeer}(s) \vee \neg \text{clown}(s) = C_6$$

$$H_6^C : \neg \text{weird}(t) \vee \neg \text{loves}(\text{Scrooge}, t) = C_7$$

$$(\neg C)^C : \text{child}(\text{Scrooge}) = C_8$$



Example 1 (contd.)

To the set of clauses $S = \{C_1, C_2, C_3, C_4, C_5, C_6, C_7, C_8\}$, **general predicate resolution** is applied. In the **resolution process**, the following **resolvents** are obtained:

$$C_9 = \text{Res}_{[x \leftarrow \text{Scrooge}]}^{\text{Pr}}(C_8, C_1) = \text{loves}(\text{Scrooge}, \text{Santa})$$

$$C_{10} = \text{Res}_{[s \leftarrow \text{Rudolf}]}^{\text{Pr}}(C_3, C_6) = \neg \text{clown}(\text{Rudolf})$$

$$C_{11} = \text{Res}_{[z \leftarrow \text{Rudolf}]}^{\text{Pr}}(C_4, C_5) = \text{weird}(\text{Rudolf}) \vee \text{clown}(\text{Rudolf})$$

$$C_{12} = \text{Res}^{\text{Pr}}(C_{10}, C_{11}) = \text{weird}(\text{Rudolf})$$

$$C_{13} = \text{Res}_{[t \leftarrow \text{Rudolf}]}^{\text{Pr}}(C_{12}, C_7) = \neg \text{loves}(\text{Scrooge}, \text{Rudolf})$$

$$C_{14} = \text{Res}_{[y \leftarrow \text{Rudolf}]}^{\text{Pr}}(C_2, C_3) = \neg \text{loves}(x, \text{Santa}) \vee \text{loves}(x, \text{Rudolf})$$

$$C_{15} = \text{Res}_{[x \leftarrow \text{Scrooge}]}^{\text{Pr}}(C_{13}, C_{14}) = \neg \text{loves}(\text{Scrooge}, \text{Santa})$$

$$C_{16} = \text{Res}^{\text{Pr}}(C_9, C_{15}) = \square$$

The **most general unifier** generated during the resolution process is the substitution:

$$[x \leftarrow \text{Scrooge}, y \leftarrow \text{Rudolf}, z \leftarrow \text{Rudolf}, s \leftarrow \text{Rudolf}, t \leftarrow \text{Rudolf}]$$

$S \vdash_{\text{Res}}^{\text{Pr}} \square$, therefore S is an inconsistent set and the deduction

$$H_1, H_2, H_3, H_4, H_5, H_6 \vdash C \text{ holds.}$$

From the hypotheses we conclude that '*Scrooge is not a child*'.

Decomposition rules for *predicate formulas*



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α - rules

$$\begin{array}{c} A \wedge B \\ | \\ A \\ | \\ B \end{array}$$

$$\begin{array}{c} \neg (A \vee B) \equiv \neg A \wedge \neg B \\ | \\ \neg A \\ | \\ \neg B \end{array}$$

$$\begin{array}{c} \neg (A \rightarrow B) \equiv A \wedge \neg B \\ | \\ A \\ | \\ \neg B \end{array}$$

β - rules

$$\begin{array}{cc} A \vee B & \\ / \quad \backslash & \\ A & B \end{array}$$

$$\begin{array}{cc} \neg (A \wedge B) \equiv \neg A \vee \neg B & \\ / \quad \backslash & \\ \neg A & \neg B \end{array}$$

$$\begin{array}{cc} A \rightarrow B \equiv \neg A \vee B & \\ / \quad \backslash & \\ \neg A & B \end{array}$$

γ - rules

$$\begin{array}{cc} (\forall x)A(x) & \neg(\exists x)A(x) \\ | & | \\ A(c_1) & \neg A(c_1) \\ | & | \\ A(c_2) & \neg A(c_2) \\ | & | \\ \dots & \dots \\ | & | \\ A(c_n) & \neg A(c_n) \\ | & | \\ (\forall x)A(x) & \neg(\exists x)A(x) \end{array}$$

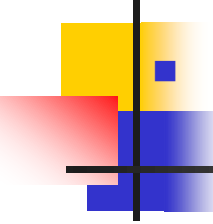
- used to decompose **universally quantified** formulas
- $c_1, c_2, \dots, c_n$ are all the parameters (constants) on that branch and they are used for the instantiation of A .
- if there is no constant on that branch, a new constant is introduced and used for instantiation.

δ - rules

$$\begin{array}{cc} (\exists x)A(x) & \neg(\forall x)A(x) \\ | & | \\ A(c) & \neg A(c) \end{array}$$

- used to decompose **existentially quantified** formulas
- c is a new constant on the branch.

Remarks

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- A formula is **inconsistent** if it is evaluated as **false** in all its interpretations. It is necessary to show that the formula does not have models, all its interpretations are anti-models.
 - To prove that a formula is **not inconsistent** it is sufficient to find a **model** for it.
 - A formula is **valid (tautology)** if it is evaluated as **true** in all its interpretations. It is necessary to show that the formula has only models. Its negation is inconsistent.
 - To prove that a formula is **not valid** it is sufficient to find an **anti-model** for it.
 - To prove that a formula is **contingent** it is sufficient to find a **model** and an **anti-model** for it.